

Database Concepts

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We store data in Files, A file can be understood as a container to store data in a computer. Files can be stored on the storage device of a computer system. Content of a file can be texts, computer program code, comma separated values(CSV), etc. Likewise pictures, audios/videos, web pages are also files.

Files stored on a computer can be accessed directly and search for desired data. But to access data of a file through software, for example to display monthly attendance report on a school website. One has to write computer program to access data from files.

Student File which has 6 columns:

RollNumber - Roll number of a student.

Sname - Name of the student.

SDateOfBirth - Date of birth of the student.

Gname - Name of the Guardian.

Gmobile - Mobile number of the students guardian.

Gaddress- Address of the guardian of the student.

Another file which is updated Regularly called Attendance which has 4 columns:

AttendanceDate - Date for which attendance was Marked.

RollNumber - The roll number of the student.

Sname - Name of the Student

AttendanceStatus - Marked as P (present) or A(absent)

Limitations of a file system

File systems become difficult to handle when number of files increases and volume of data also grows.

Difficulty in Access: Files themselves do not provide any mechanism to retrieve data. Data maintained in a file system are accessed through application programs. While writing such programs, the developer may not anticipate all the possible ways in which data may be accessed. So sometimes it is difficult to access data in the required format and one has to write application program to access data.

Data Redundancy: Redundancy means same data are duplicated in different places (files) In our example, student names are maintained in both the files. Students may have. Same guardian name and therefore same guardian name is maintained twice. This is the example of redundancy which is difficult to avoid in a file system. Redundancy leads to excess storage use and may cause **data inconsistency** also.

Data inconsistency: Data inconsistency occurs when same data maintained in different places do not match. If a student wants to get changed the spelling of her name. It needs to be changed in Sname column in both the files. Likewise, if a student leaves school, the details needs to be deleted from both the files. As the files are being maintained by different people, the changes may not happen in one of the files. In that case the student name will be different (inconsistent) in both the files.

Data Isolation: Both the files student and attendance are related to students. But there is no link or mapping between them. The school will have to write separate programs to access these two files. This is because data mapping is not supported in file system. In a more complex system. Data files are generated by different persons at different times. Files being created in isolation may be of different formats. In such case it is difficult to write new application program to retrieve data from different files maintained at multiple places as one has to understand the underlying structure of each file as well.

Data Dependence: Data are stored in a Format or structure in a file. If the structure or format itself is changed, all the existing application programs accessing the file also needs to be changed. Otherwise the program may not work correctly. This is data dependency. Hence, updating the structure of a data file requires modification in all the application program accessing that file.

Controlled Data Sharing: There can be different categories. Categories of users like teacher, office, staff and parents. Ideally, not every user should be able to access all the data. As an example, guardian and office staff can only see the student attendance data, but should not be able to modify or delete it. It means these users should be given limited access (read only) to the attendance file. Only the teacher should be able to update the attendance data. It is very difficult to enforce this kind of access control in a file system where accessing file through application programs.

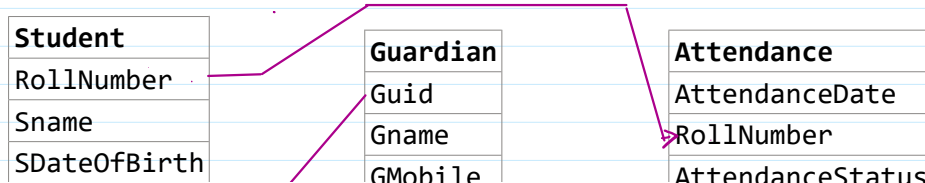
DATABASE MANAGEMENT SYSTEM

Limitation faced in the file system can be overcome by storing the data in a database where data are logically related. We can organize related data in a database so that it can be managed in an efficient and easy way.

A database management system (DBMS) or database system in short is a software that can be used to create and manage databases. DBMS let's user to create a database store, manage, update, or modify and retrieve data from the database by users or application programs. Some examples of open source and commercial DBMS include MySQL, Oracle PostgreSQL, , Microsoft SQL server, MongoDB, DB2 by IBM etc.

A database system hides certain details about how data are actually stored and maintained. Thus it provides users with an abstract view of the data. A database system has a set of programs through which users or other programs can access, modify, and retrieve the stored data.

The DBMS serves as an interface between the database and end users or application programs. Retrieving data from a database through special type of command is called querying the database. In addition, user can modify the structure of the database itself through a DBMS.



RollNumber	Guid	AttendanceDate
Sname	Gname	RollNumber
SDateOfBirth	GMobile	AttendanceStatus
Guid	GAddress	

Key Concepts in DBMS

In order to efficiently manage data using a DBMS:

Database Schema: Database schema is the Design of a database. It is the skeleton of a database that represents the structure (table names and their fields or columns) The type of data each column can hold. Constraints on the data to be stored (if any) And the relationship among the tables. Database schema is also called the visual or logical architecture as it tells how the data are organized in a database.

